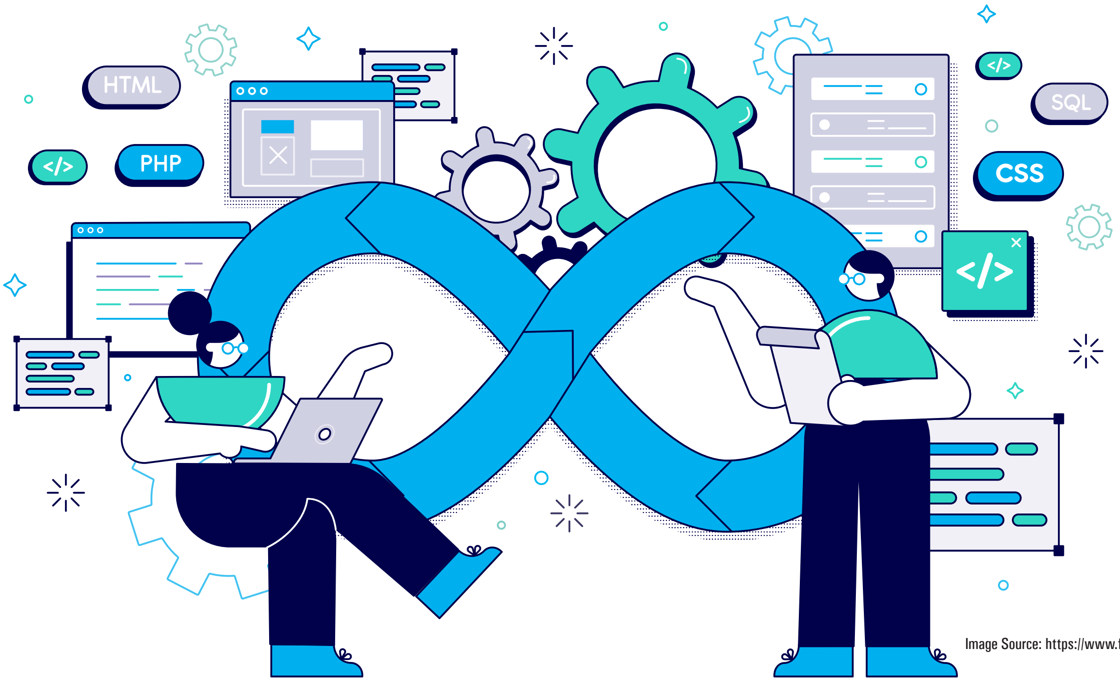
# Integrating Network Function Virtualization with the DevOps Pipeline: Distributed Systems

This multi-part series of articles will take you through the various components of DevOps. It will introduce you to network function virtualization and its integration with the DevOps pipeline. This first article in the series focuses on distributed systems.

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Service providers around the globe are looking for concrete solutions to provide the latest telecom applications and comply with the demands of their users. Proprietary networking devices are used to deploy these applications. While this solution fulfils customer requirements, it is not scalable and demands hefty financial investment because such hardware can easily get outdated with new technologies emerging in the market.

Network function virtualization (NFV) is seen as something that brings service providers out of this bottleneck. It is seen as a cost-effective implementation of telecommunications infrastructure. This series of articles will deal with the integration of NFV modules in a DevOps pipeline, while showcasing how various DevOps tools are interconnected to produce an environment capable of handling and deploying an NFV based infrastructure on an open source cloud.

With the advent of DevOps tools and methodologies, we now have a system for NFV applications that can be deployed and delivered in a jiffy. The various methodologies of DevOps fulfil the requirement of fast-paced delivery with easy-to-configure servers and no downtime of the application. NFV application, when orchestrated by the Kubernetes cluster, enables independent and dynamic scaling. NFV system enablement reduces the CAPEX and